Computational Pipeline Choice Drives Shotgun-Amplicon Discordance in Microbial Community Profiling: A Fisher-Rao Distance Approach

Clark Gaylord Marcos Pérez-Losada Keith A. Crandall
Dept. of Biostatistics & Bioinformatics and Computational Biology Institute
GWSPH Research Day, April 15, 2026

GWSPH Research Day

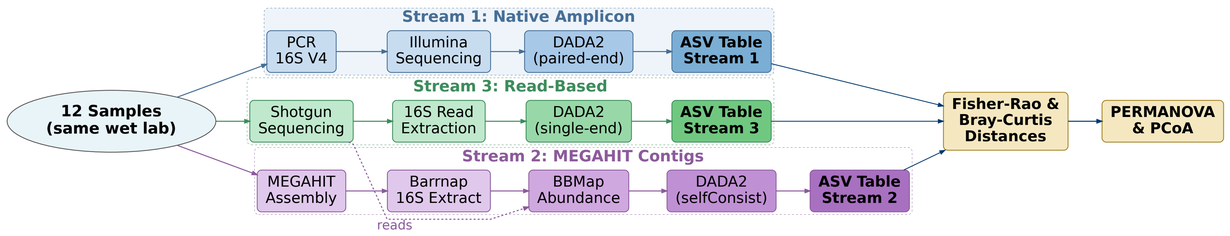
GW Public Health

Background & Objectives

Shotgun metagenomics and amplicon sequencing frequently yield discordant community profiles, commonly attributed to biological or wet-lab differences — but how much arises from *computational pipeline* choices?

We isolate the dry-lab contribution by routing shotgun data through the amplicon pipeline, holding wet-lab inputs constant. We ask whether Fisher-Rao distance detects pipeline-driven differences more reliably than Bray-Curtis.

Dataset: Pérez-Losada et al. (2022) vermicomposting; 12 microcosms, 2 × 2 factorial (earthworm × time); 3 replicates per cell. 554 ASVs; 99.1% phylum assignment (SILVA 138.1).



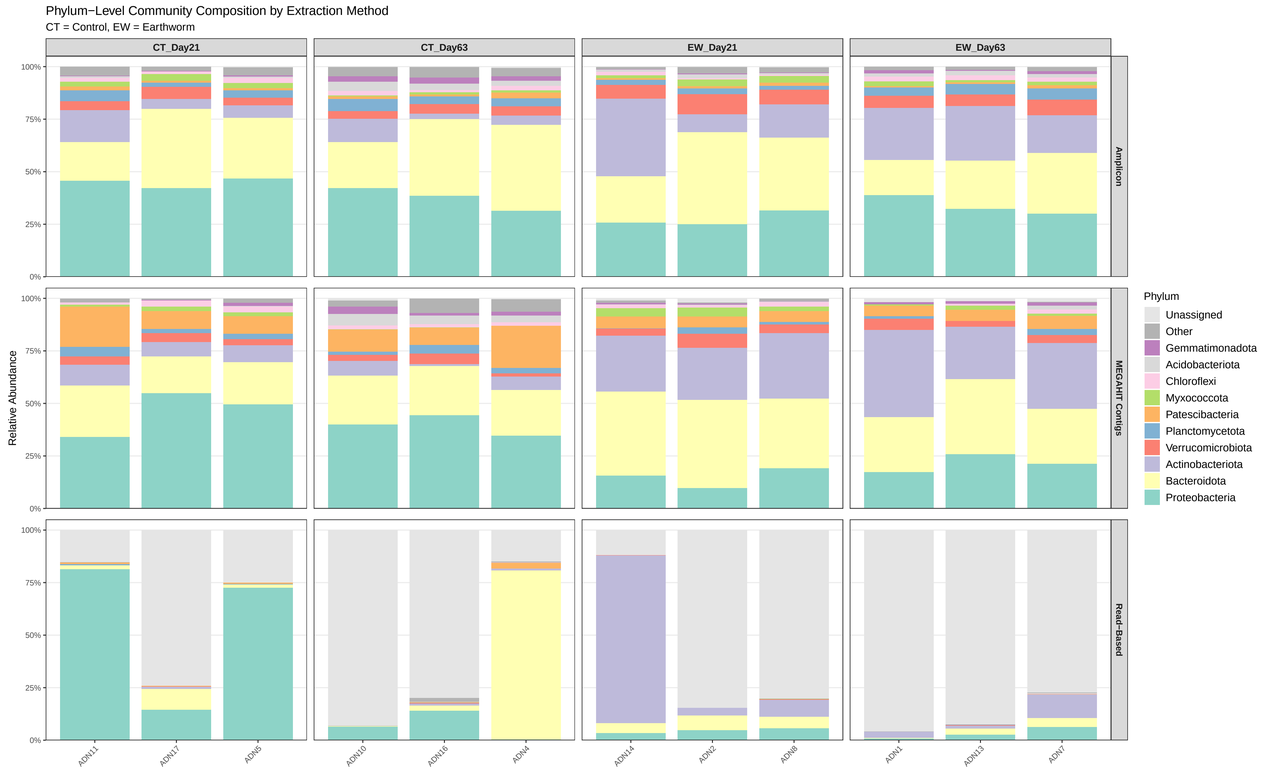
Stream 1 — Native Amplicon: 16S V4 PCR → Illumina → DADA2.

Stream 2 — MEGAHIT Contigs: Shotgun → MEGAHIT → Barrnap → BBMap → DADA2.

Stream 3 — Read-Based: Shotgun reads extracted and routed through amplicon DADA2.

Streams 2 & 3 share identical wet-lab inputs, isolating computational pipeline as the variable of interest.

Phylum-Level Composition by Treatment × Time



Rows: analytical stream. Columns: experimental group (CT/EW × Day 21/63), 3 replicates per cell. Amplicon and MEGAHIT Contigs show concordant treatment response; Read-Based diverges (<0.1% of shotgun reads match V4). FR distances below quantify the difference.

Fisher-Rao Distance & PERMANOVA Results

Fisher-Rao distance arises necessarily from the Fisher Information Matrix and is the *unique* metric on the probability simplex invariant under sufficient statistics (Čencov, 1982). Bray-Curtis was adopted by historical convention, not derived from the multinomial model.

For $\mathbf{p}$, $\mathbf{q}$ over K taxa:

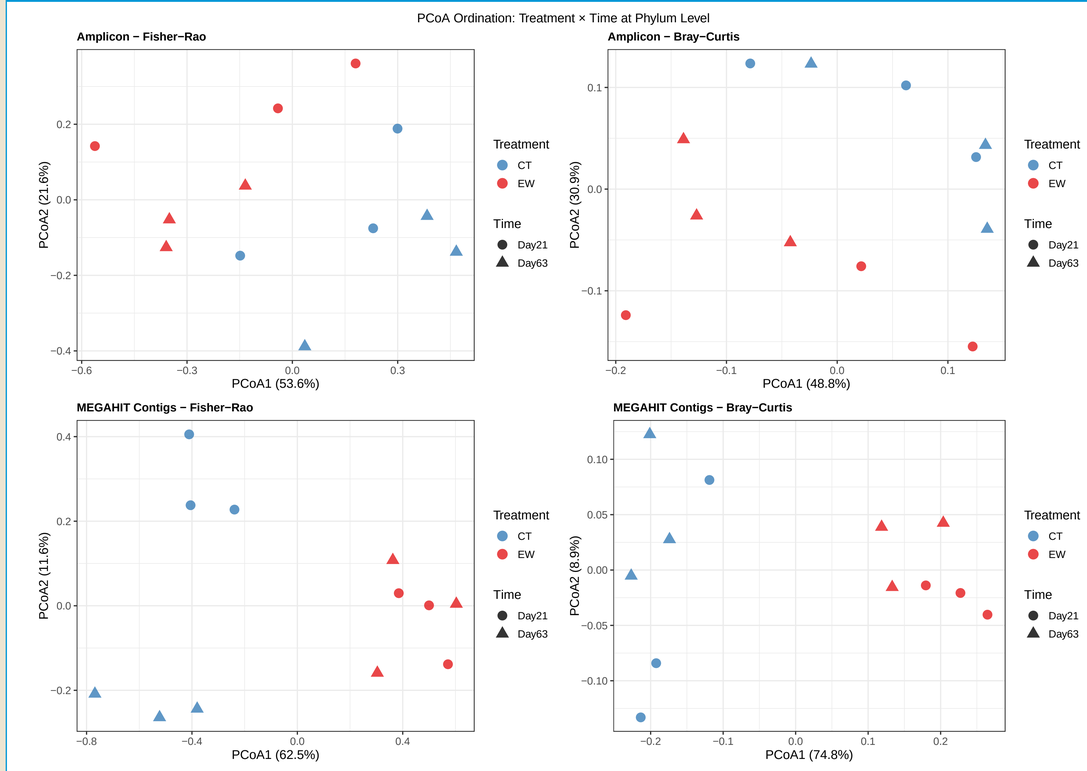
$$D_{\mathrm{FR}} = 2\sqrt{K}\arccos\left(\sum_{k=1}^{K}\sqrt{p_k q_k}\right)$$

PERMANOVA (9,999 permutations, phylum level):

Stream	Effect	*p*-value FR	*p*-value BC
Amplicon	Treatment	0.004**	0.007**
	Time	**0.073.**	0.190
MEGAHIT	Treatment	0.0001***	0.0004***
	Time	0.079.	0.093.

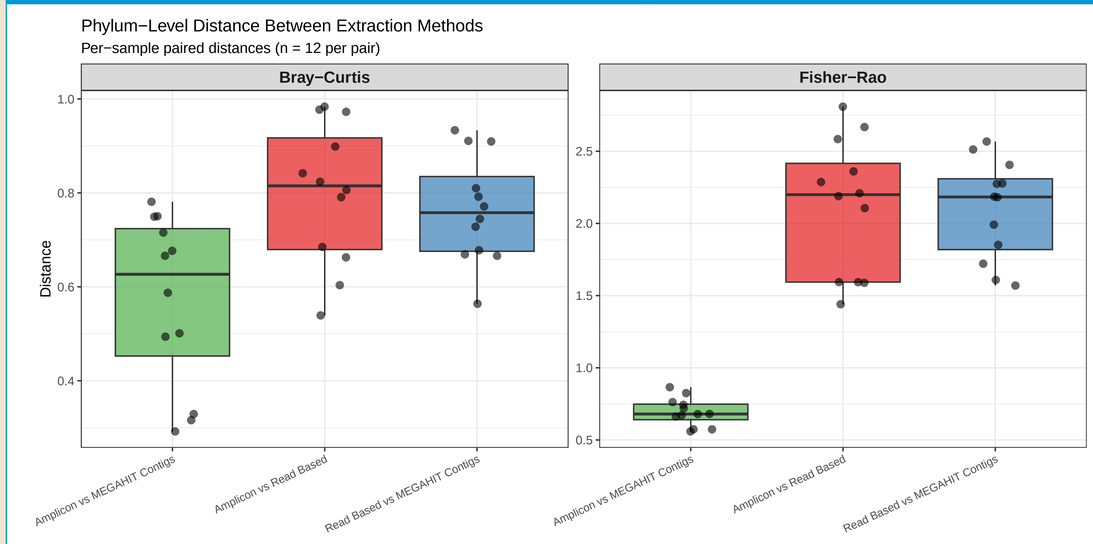
FR detects time effect ($p = 0.073$) missed by BC.

PCoA Ordination by Treatment × Time



Fisher-Rao (left) and Bray-Curtis (right) for Amplicon and MEGAHIT Contigs. Treatment and time structure visible under both; Fisher-Rao gives tighter clustering.

Pairwise Distances: Fisher-Rao vs. Bray-Curtis



FR confirms concordance ($\bar{d} = 0.693$) and divergence (3× larger for Read-Based), with lower variance than BC (CV ratio ≈ 0.44) — consistent with the composition above.

Conclusions & Future Directions

Much reported shotgun-amplicon discordance reflects *dry-lab pipeline differences*.

Routing shotgun reads through the amplicon pipeline recovers concordant community profiles.

Fisher-Rao distance, derived necessarily from the Fisher Information Matrix of the multinomial distribution, provides more stable community comparisons than Bray-Curtis.

Further study of the sampling distribution of Fisher-Rao distance will support power analysis and Type I error characterization for PERMANOVA-based testing.

Validation against mock communities with known composition will establish the accuracy of contig-based 16S extraction across sequencing platforms.

Data: Pérez-Losada et al. (2022), *Front. Microbiol.* 13:854423. BioProject PRJNA777435.

Computing: Pegasus HPC cluster, GW Research Technology Services.

Theory: Čencov (1982), *Statistical Decision Rules and Optimal Inference*; Rao (1945).

Manuscript: Claude (Anthropic).

Poster & Draft

github.com/cgaylord/2026GWSPH_ResearchDayPoster